Microscopy CoRE – Leica Stellaris Basic Operation (Confocal)

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The Leica Stellaris 8 microscope is a confocal imaging system designed to maximize the number of probes and photon collection sensitivity for a wide range of applications, including fixed and live samples. The Stellaris 8 is equipped with a number of technologies that allow it to achieve this versatility and sensitivity. A White Light Laser (WLL) can pump 8 individually tunable and simultaneous laser lines, and it is tunable from 440 nm to 790 nm in 1nm steps. The system features an additional UV405 diode laser bringing the total usable lasers to 9 simultaneously. The system is equipped with 4 spectral PowerHyD detectors having a detection range from 405nm to 850 nm, a photon sensitivity of up to 58%, and a dynamic range up to 180MCounts/s. The spectral capabilities of the Stellaris 8 are augmented by the TauSense technology. TauSense allows real-time separation of fluorophores based on their lifetime properties. It achieves this by computing the distribution of the average arrival time (AAT) of photons. The user is then able to separate fluorophores or remove auto-fluorescence based on these lifetime properties. Tiling and imaging multi-ROI samples is easily achieved in the Navigator interface, and an Okolab incubation box controls temperature, humidity and CO2.

The Stellaris 8 system is also equipped with Lightning Super-Resolution, a deconvolution-based strategy (Richardson-Lucie algorithm coupled to a statistical decision mask that computes SNR in voxels of data), that can improve resolution to 120 nm in the lateral dimension and 200 nm in the axial dimension. The algorithm can run in a global or adaptive setting depending on the user selection, while preserving the sum of photons in the volume and raw confocal data and runs in near-real-time on up 5 simultaneous detection channels without the need for calibration.

Prerequisites

1. **Before and after your session, you are responsible for cleaning the objectives.**
    a. Turn on the microscope according to the System Startup below.
    b. Open the doors to the incubation box. *Do not loosen the top screws all the way or they will fall out.*
    c. Use the touchscreen to switch between objectives (Objective Lens icon).
        i. **Immersion lenses** - Use lens paper and lens cleaner.
            1. Tear off one sheet of lens paper from the packet. Wet a small area of lens paper with the cleaner (pink liquid). **Do not use KimWipes** to clean any microscope optics.
            2. With your pointer finger, gently press the wet area to the front lens of the lens and wipe in a circular motion 1 or 2 times.
            3. Switch to a dry area of the same paper and gently wipe 1-2 times in a circular motion.
            4. Repeat as necessary until the lens is clean. Use fresh lens paper and cleaner for each lens; use fresh paper as well if you must wipe off a lens more than once.
        ii. **Dry/Air lens** – Use lens cleaner, lens paper and cotton swabs to clean.
            1. Soak one cotton-tipped swab with lens cleaner.
            2. Gently wipe the front of the lens in a circular motion.
            3. Place the lens paper over the lens. Using a second dry cotton swab, gently rub in a circular motion to dry off the lens.
            4. Repeat as necessary until the lens is clean. Use fresh lens paper and cleaner for each lens; use fresh paper as well if you must wipe off a lens more than once.
2. **Before imaging, be certain that any mountants or sealants are completely dry.** This typically takes at least 24 hours from the time the slide is prepared. Refer also to manufacturer's instructions.

Do not use KimWipes to clean any microscope optics. KimWipes are for slides only!

System Overview

System Startup

1. At left from the system, black box, flip on the two switches.
2. Under to the switches, turn the laser key clockwise to the ON position.
   Ensure the "emission" light glows white
3. Log on to the computer using your credentials that you set up during the training.

1. Power switch
2. Laser switch
3. Turn Emission Key to vertical.

Do not turn the key on the White Light Laser – it should always be pointed toward ARMED.

4. Log on to the computer with the credentials you entered during your training.

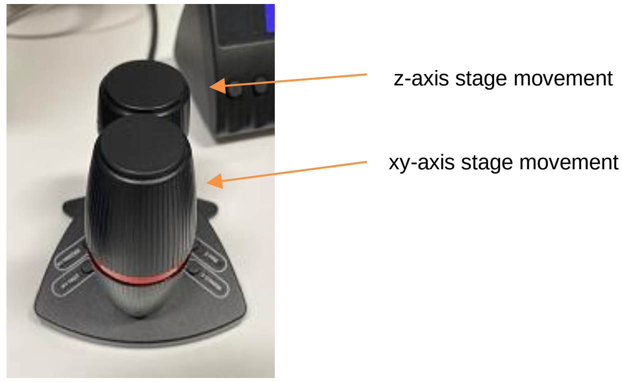
Basic Fluorescence Setup

1. On the microscope, gently tilt the transmitted light arm to the rear until it stops.

2. There are three sample holders available for this system. *The slide holder is typically installed. If you need the well plate holder or the incubation holder, please alert Core staff when booking the instrument.* ***Do not push too hard on either sample holder due to their delicate nature.***

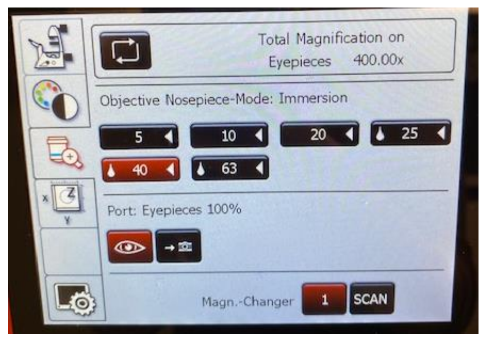
   a. If you are using slides, place the slide coverglass face down in the center of the slide holder. Move the slide clips over the slide to secure it in place.

3. Gently tilt the transmitted light arm back into place over the stage. Center your specimen using the XY SmartMove controller over the objective lens.



4. *On the touchscreen* – Along the left edge, tap the **Lens Control** tab (objective lens with a "+" symbol).

   Tap the 10x objective (twice if it does not move right away).



5. *On the touchscreen* – Along the left edge, tap the **Filter Control** (colorwheel/black-white wheel). Along the top edge, tap **FLUO** under "Incident." Alternatively, focus your sample using brightfield illumination by tapping **BF** ("Transmitted").

   Along the bottom edge, select one of the three available widefield filters – DAPI (blue), FITC (green), or RHOD (red).

6. *On the touchscreen* – Tap **IL Shutter**. Fluorescent light should now appear on the stage. *A yellow circle will appear next to "IL Shutter" on the screen.*

7. Use both hands to set the correct interpupillary distance: when you look through the eyepieces, you should see one image circle instead of two.

*If you do not see light through the eyepieces, briefly return to the Lens Control page and tap the "eye" icon under **Port** (see screenshot in Step 4).*

8. Use the XYZ controller to center and focus on your sample.

9. *On the touchscreen* – Tap either of the two remaining widefield filters to ensure there is signal and specificity for all fluorophores used in your sample.

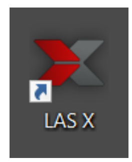
If your sample does not appear to be exhibiting fluorescent signal where you would expect, or as brightly as you may expect, this may be a good time to return to the lab to review your sample preparation methods to see if anything can be adjusted for a better result.

10. When initial observation is complete, tap **IL Shutter** (Filter Control) again to turn off the fluorescence lamp.

11. To switch to an oil immersion lens –Tap on the Lens Control tab. Tap on the required magnification TWICE – tap once, wait for the lens to flash, then tap again. The oil immersion lens will swing into place. Carefully tilt the transmitted light arm back and remove your sample. Apply one drop of immersion oil using the paddle on to the coverglass of your slide. Replace your sample and refocus. The oil should make clear contact with the lens.

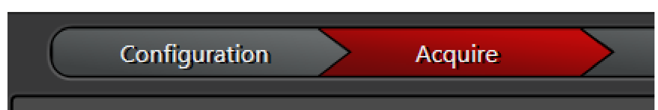
Software Startup

1. On the desktop, launch **LAS X**.



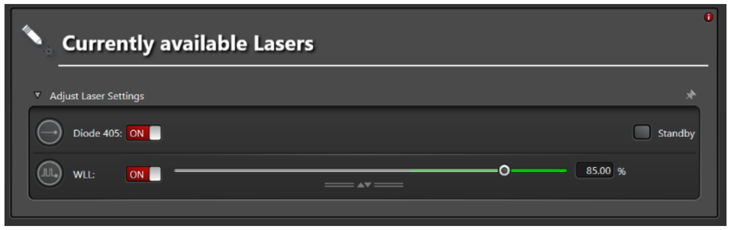
2. When prompted, set the options as follows and click OK.

3. Switch to **Configuration** along the top edge.

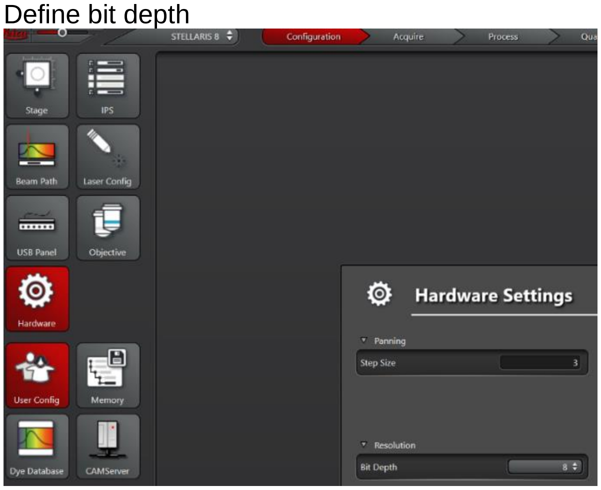


4. Click on **Laser Config** along the left edge. Set the following percentages for activated laser lines. *Only activate lasers that will be needed for that experiment – leave others turned off.*

Diode 405 – for DAPI, DyLight 405, etc.
WLL – All other fluorophores with excitation spectra at 440nm and longer. Available range: 440-790nm.



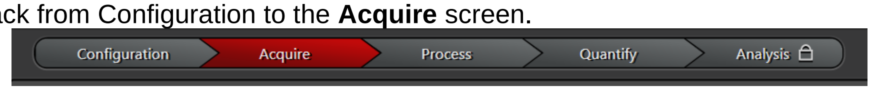
5. Define bit depth



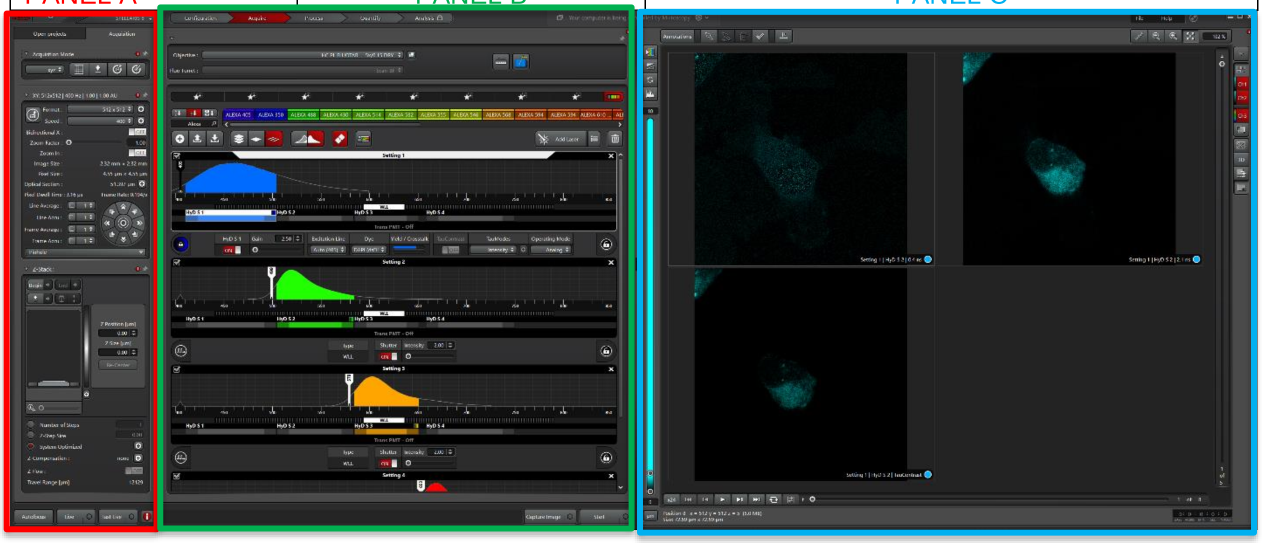
Most people only need 8-bit depth for their confocal imaging need. However, sometimes you need more depth because your sample has very bright and very dim but equally important signal, or you would like accurate quantitative analysis. Go to Settings (Top of screen) and select the Hardware tab Select the appropriate bit depth for your image: 8-, 12- or 16-bits

Imaging Setup (Software)

1. Switch back from Configuration to the **Acquire** screen.



PANEL A	PANEL B	PANEL C



2. In **Panel A: Scan Parameters**, set **Format** to 1024 x 1024 and **Speed** to 600 Hz

3. You can define you light settings in **Panel B: Light Path** by using the Dye Assistant, manually or loading existing settings from the image collected last time.

    a. Dye Assistant 

    Click on the Dye Assistant icon at the top of the Light Path panel and new window will pops up.

    Click on the "…" icon and find your dye choice. Enter as many dyes as you need. The software will give you a series of combinations of light settings and asks you to select the one you would like.

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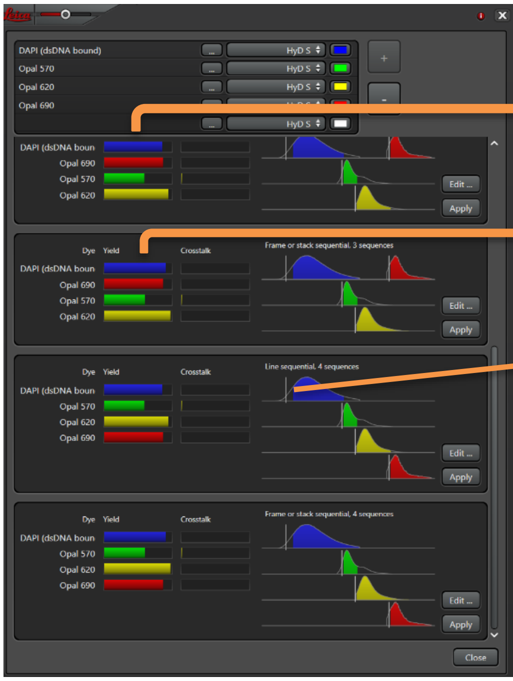


Settings have three informative sections:

On the left is the yield for each dye, it tells you how much light from your dye is collected.

In the middle is crosstalk, which tell you how much light from a dye will be detected when collecting light from a different dye.

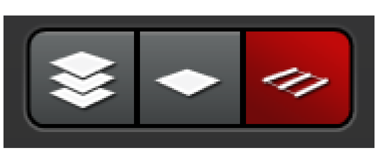
On the right are the emission spectra of your dyes, the white vertical bars represent the lasers.

Select the settings that are appropriate for you and click “Apply”.

- **None sequential**: fastest mode for acquisition. All fluorophores are detected at the same time using different detectors. Since all the lasers needed to excite the fluorophores are “on” at the same time, this can lead to signal crosstalk or bleed-through (detection of two different fluorophores in the same channel) --> set all lasers and detectors together.

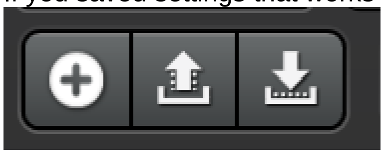
- **Line sequential**: to avoid crosstalk, the excitation-detection of the different fluorophores can be alternated. In this case, the excitation-detection of the different fluorophores switched between lines until the whole frame is scanned. This mode is optimal for scanning fast events since the switch between sequences is very fast. In this mode, the hardware settings (detector position, averaging, accumulation…) need to be the same for all sequences need to be the same to allow the fast switching --> click SEQ in the Acquisition mode. Set the laser/s and detector/s for each sequence. Choose “Between Lines” in the Sequential Scan

- **Frame sequential**: similar to line sequential but the switch between sequences is done after acquiring the whole frame. For this mode, hardware settings can be adjusted individually for each channel giving more flexibility for the detection of each fluorophore --> click SEQ in the Acquisition mode. Set the laser/s and detector/s for each sequence. Choose “Between Frames” in the Sequential Scan

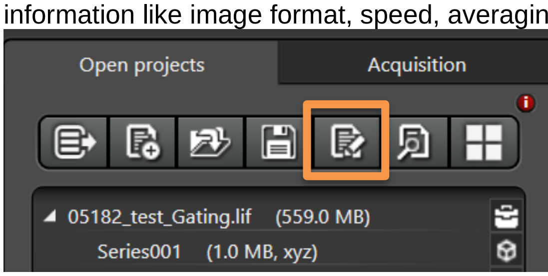
 You can switch between Stacks/ Frames/ Lines by clicking on the icon above the Tracks settings.

b. Loading existing settings.

If you saved settings that works for you can load them by clicking the button “+”



Alternatively, you can open a file with images and select the Apply button/ right click on the image and Apply image settings. (Apply copies only light setting and laser power. It doesn’t copy any other information like image format, speed, averaging, accumulation etc.)



Before you start imaging review your settings one by one, for example if the do not interfere with one another.

Image Acquisition

1. In the bottom left of the screen, click “Live” to start your preview scan.

2. You will see an image on the right-hand side of the screen and on its top left a series of controls. Click the Over/Under exposure: it will show you absence of light in green and saturation in blue.

3. Set your gain and laser power gradually until you get an image with as little background as possible and no saturation (Try not to go over 100% on the Gain). Use the z-position knob to find your brightest focal plane in your specimen of interest. It’s worth considering adjusting your setting on the brightest sample.

4. Click “Stop” in the bottom left (“Live” button) to stop your live scan.

5. If different channels are being acquired. Click on the next sequence and repeat steps 1-4 for each channel.

6. If signal is still low, adjustments on “Line accumulation” might be needed.

7. Adjust format/speed again if needed. Additionally, line/frame average can be increased to reduce background.

8. Click “Start” (bottom) to capture your image (all the channels). Click “Capture image” to only acquire the active sequence.

9. To save your image/s, click on “Open projects” (tab next to “Acquisition” parameters):

10. Save your images on the Hive in the MicroscopyPublic folder (Map the Drive at the beginning of each session).

Defining z-stack/3D image

Once your settings are defined, you may want to define a volume to be scanned – a stack. This is found if you are scanning in XYZ and the controls are on the middle left hand side of your screen.

1. Go “Live”
2. Use the Z-position on the panel (first on the right), turn it clockwise until your signal of interest disappears or wherever you want to start your z-stack and click on “Begin”, it will turn from grey to red.
3. Turn opposite direction until your signal of interest disappears or wherever you want to end your z-stack and click “End”, it will turn from grey to red.
4. To go to the beginning or end of your stack, click on either the arrow to the right of Begin or End. To go to the middle of the stack, click on the symbol right of the bin. To delete the stack settings, click on the bin. You have a choice to define the number of steps, the thickness of steps or to let the system use optimized step size. We recommend that you use the later if you want to accurately reconstruct your volumes.

The Navigator Module

The Navigator Module you can access by clicking on the square grid under Acquisition. Before using it, it’s worth programming your light settings in normal mode, defining your stack, etc… Move to the middle of your stack if you have one.

Clicking on the grid opens a new window. All the control panels you accessed previously are arranged in a series of tab on the left of the screen. Click on the “scanmode” icon, you can change for smaller format image and smaller zoom to scan spiral overview faster.

1. Making a map: Spiral mode.

   You can quickly see your sample by clicking “Live” and you’ll see a single slice taken using a single light setting. For a wider view of your sample, click on “Spiral” and the software will make a spiral, taking a single slice using all channels at every field of view. Once you have seen enough, click “stop”. You can move with objective to different part of your sample, by double clicking and start the spiral in a new position.

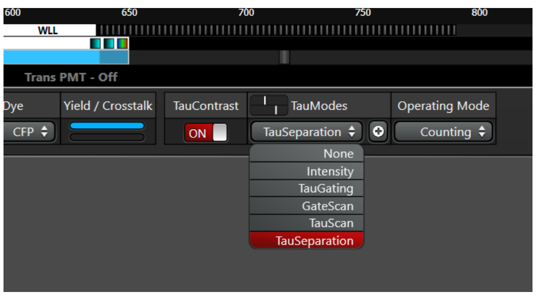
2. Defining the region of interest.

You can use the map you have made to choose your region of interest by simply clicking on it and the microscope will automatically center its field of view where you have clicked. You can even change lens and -provided you don't move your sample- the microscope will adapt its field of view. Remember that you may need to adapt your light settings and image format to the lenses Should your region of interest be larger than the field of view, you can use the navigator buttons to draw a region and the computer will define the number of fields of view you need.

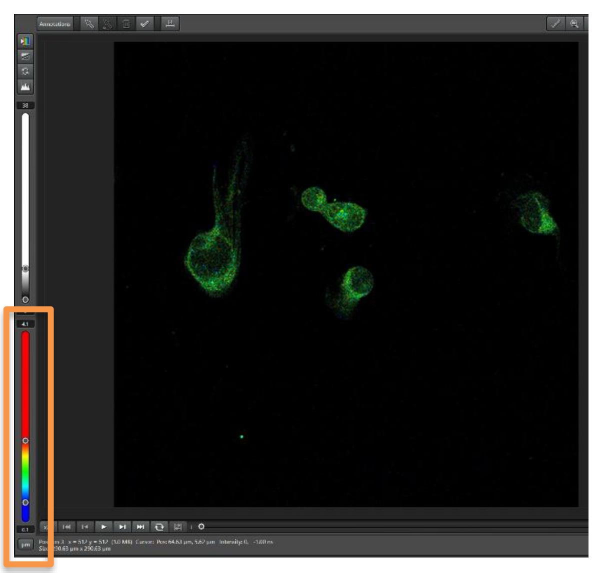
3. Defining several regions of interest.
   It is possible to define several regions on interest at the same time. These may however have different depth. Click on Z-Stack. You'll find these options available You can then define the Z range as you define each region of interest

Tau Sense

TauSense technology is a set of imaging tools based on fluorescence lifetime. It gives you the option of visualizing the fluorescence lifetimes in your specimen at the same time that the fluorescence intensities are visualized.



To do so, various gating and Tau modes for separating signals are at your disposal in the TauModes area.



Tau Contrast- This function determines the average photon arrival time per pixel in the detector and displays it as a color lifetime image in a separate channel. By adjusting the histogram, you can explore average lifetimes of various structures.
*provide immediate access to functional information, such as metabolic status, pH, an ion concentration

None- Only if the TauContrast function is switched on. Displays only the lifetime image and not an intensity image of the specimen.

Intensity- Without gating. Displays the full intensity image of the specimen.

TauGating- enables splitting photons arriving at different times. Allows the positioning of multiple digital gates (up to 16) with precise and flexible time definitions
*removes unwanted fluorescence contributions ex. Autofluorescence
* TauGating delivers images that contain the number of photons (intensity) detected in one gate (or a set of gates, if several gates are pooled) during the pixel dwell time.

GateScan- For each channel, time gates of different lengths but with identical end times are scanned.

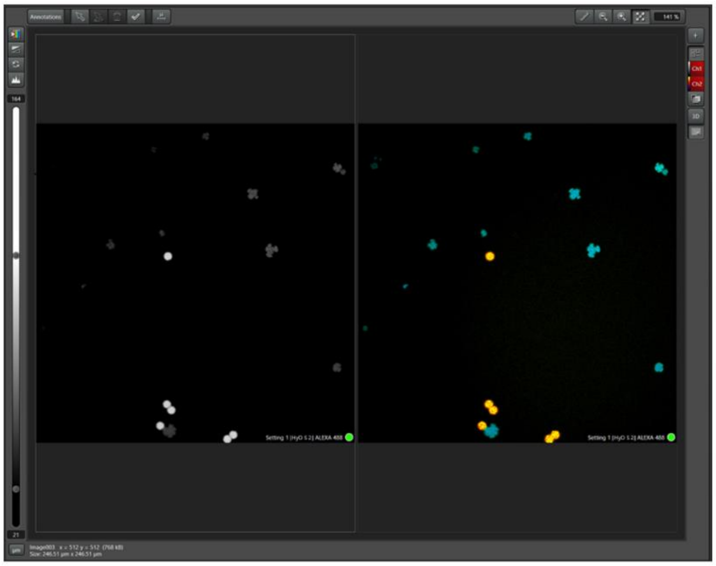
TauScan- Using the Tau scan, you can obtain a spatially resolved lifetime diversity curve. With the aid of the diagram, the lifetime components in the different species of the specimen can be determined. In this way, you can also determine whether a dye contains more than one lifetime component, for example.

TauSeparation- With this method, you can carry out a separation (e.g. dye separation) of species present in the specimen based on different fluorescence lifetimes that you have determined, e.g. in a preceding Tau scan. For each lifetime component, an intensity image is generated in a separate channel and displayed in a viewer partition.

Separation of species present in the specimen using different fluorescence lifetimes (t) provides an additional dimension for separation of species.
This method is suitable e.g. for specimens that have been stained with multiple dyes whose emission spectra overlap but significantly differ in the fluorescence lifetimes (e.g. mRFP and mCherry).

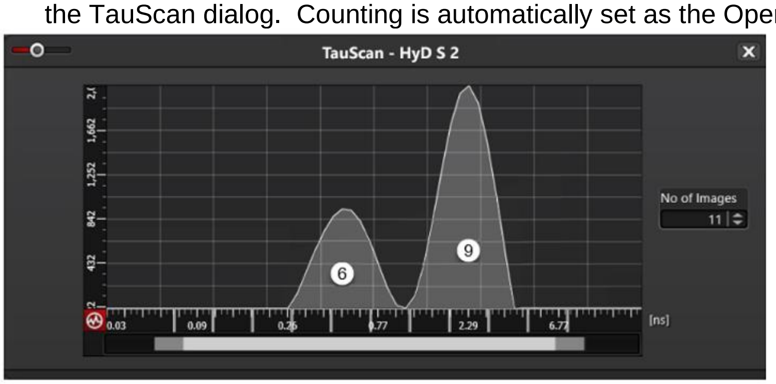
The following instructions guide you through the individual procedural steps to separate species based on different fluorescence lifetimes.

1. Turn ON TauContrast. The first step tests whether a species contains multiple lifetime components. The TauContrast function is used to measure the average photon arrival time per pixel in the detector and display it as a colored lifetime image. Two images are acquired per detector and displayed in the viewer: The first image the fluorescence intensity is shown through the brightness and the second image additionally uses colors to represent the average arrival times
2. Click START



Two images are created per setting and displayed in the viewer: One monochrome intensity image (left) and one colored lifetime image (right).
If multiple species can be distinguished by color in the lifetime image, this is an indication that they have different lifetimes. Two different species can be identified in the example shown: Cauliflower-like blue structures and round yellow structures.
In the Quantify operating step, you can use the lifetime image to carry out the same analyses as with an intensity image.

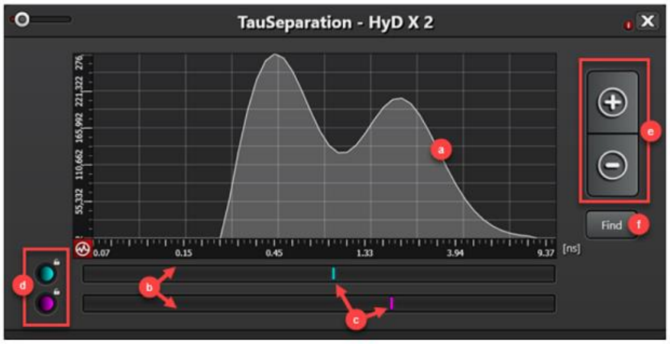
3. Switch the TauContrast function off.
4. Next, you can use a Tau scan to identify the lifetime components for the different species in your specimen.
5. **TauScan mode.** In this step, lifetime components are assigned to the different species. A Tau scan gives you a diagram of the frequencies of fluorescence lifetime components, also referred to as a lifetime diversity curve. Here, you can combine the detectable lifetime components into ranges. One channel is assigned to each range and the corresponding intensity image for each is displayed in the viewer in a separate partition. Each partition shows information about the setting, detector and corresponding lifetime. Accordingly, the different species in the specimen become visible and can be assigned based on their individual lifetimes.
6. Switch on LIVE. Select the detection channel in which you want to carry out the Tau scan and disable all other settings.
7. Under TauModes, select the TauScan entry for the selected detector and click the "+" icon to open the TauScan dialog. Counting is automatically set as the Operating Mode.



Use the time scale under Nr. of Images to set the number of lifetime ranges into which the entire detection range is to be divided, or – in other words – how many images are to be created.
Now a diagram is displayed in the TauScan dialog. The curve displays how frequently (counts) which lifetime components appear in the entire image. Each peak indicates a lifetime component present in the specimen (Sections 6 and 9).

8. Each lifetime range is represented in the viewer by a partition in a gallery display. Each partition shows an intensity image of the corresponding lifetime range. One color field is displayed for each channel. The color corresponds to the LUT configured in each case. Based on the diagram and the visualization in the viewer, you can identify which species contains which lifetime components and how you can best assign the lifetime components to different species or channels.
9. Now you can separate these based on their lifetimes using the TauSeparation method.

10. **Tau separation mode.** This method separates the species and acquire separate images.
11. Change to the TauSeparation method, and click the "+" icon to open the TauSeparation dialog and switch the LIVE again.



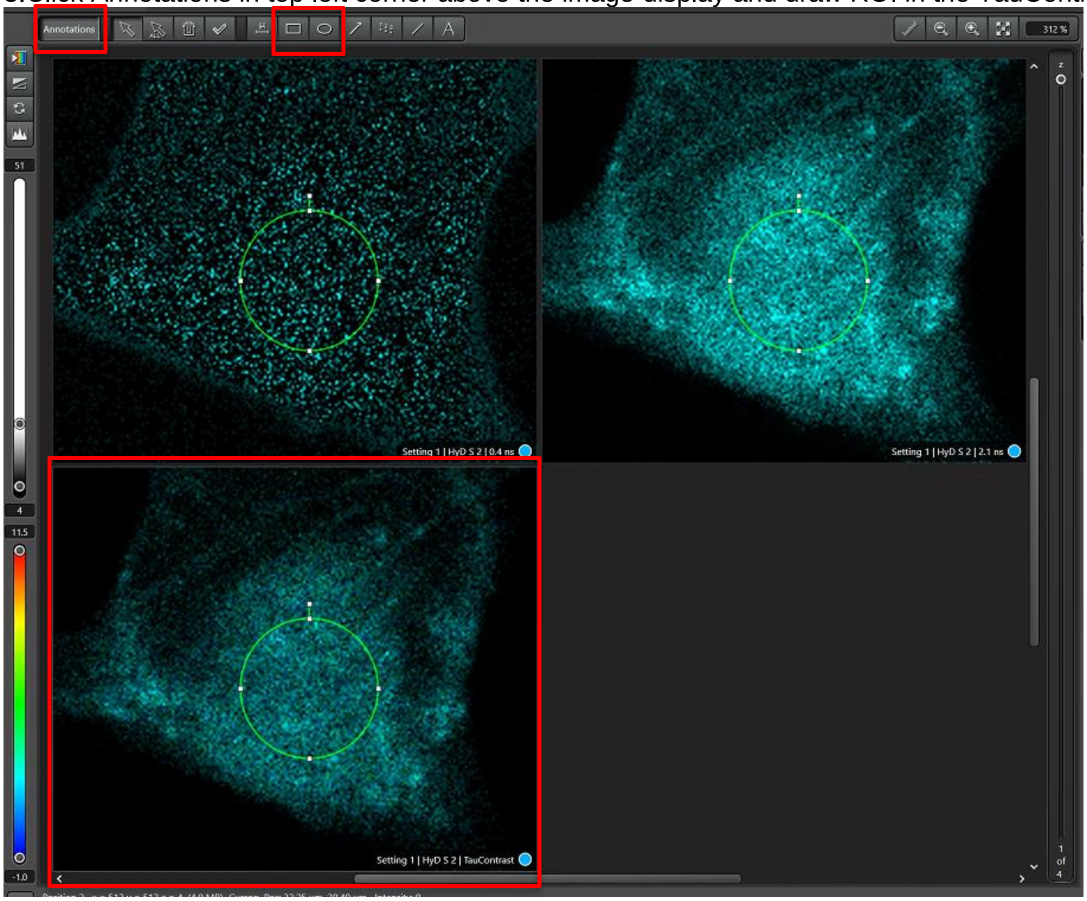
The diagram (≙ TauScan) of lifetime components detected in your specimen is displayed here (a). The diagram shows the total number of photons counted in the image over the lifetime (time difference after the laser pulse in nanoseconds). Each peak represents a lifetime component present in the specimen and is assigned to a separate channel (b).

12. Identify the number of components (= number of peaks) based on the lifetime distribution curve (a). There are two in the example.
13. Assign the lifetime components to the different species based on the peaks in the lifetime diversity curve (a) and the visualization of the Tauscan in the previous step.
    To do so, proceed as follows:
    - Move the lifetime lines (c) under the peaks if that has not already been done automatically.
    - Use the + button to add more channels (e) until all identified species are represented by a line. By clicking -, you can also remove the corresponding channel again.
    - Click the Find button (f) to start the analysis of all channels.
14. Click the Start button to start the separation of the species present in the specimen. The calculation is performed, the species are automatically separated and a channel is assigned to each species. One intensity image is created for each channel and displayed in the viewer.

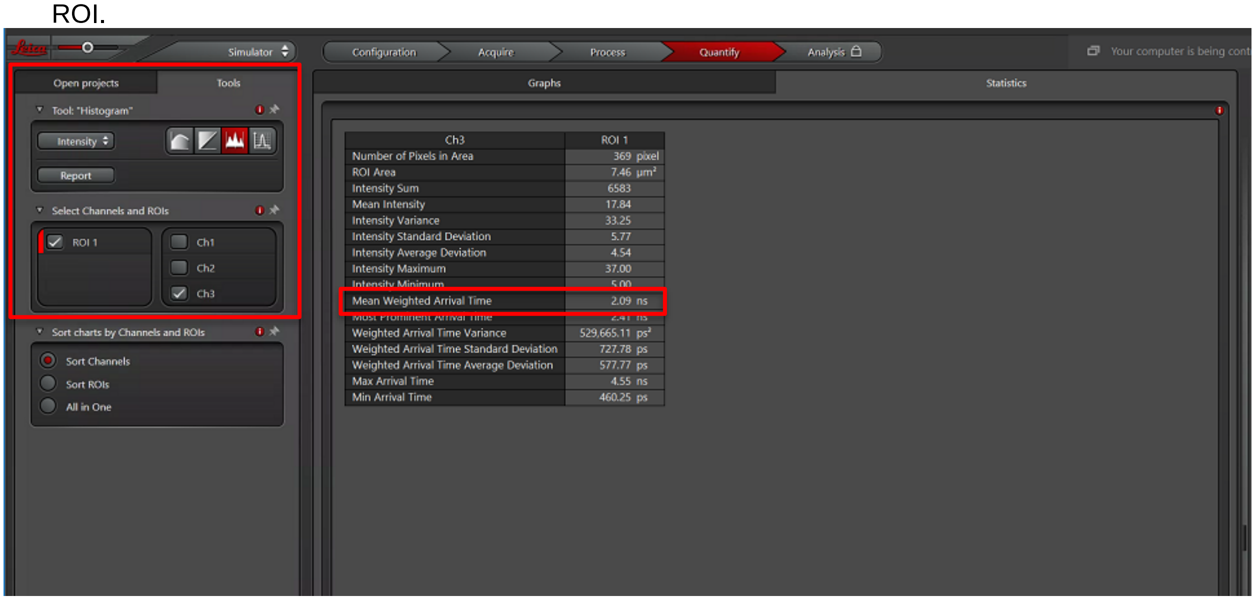
Average Arrival Time (AAT) data export

1. Open the image that you acquired with TauContrast.
2. Go to Quantify tab at the top

3.Click Annotations in top left corner above the image display and draw ROI in the TauContrast window.



4. Navigate to Tools tab and click on Histogram. You can select ROI only in the TauContrast channel (ROI3). In the table under Statistics tab you can find Mean Weighted Arrival time of your ROI.



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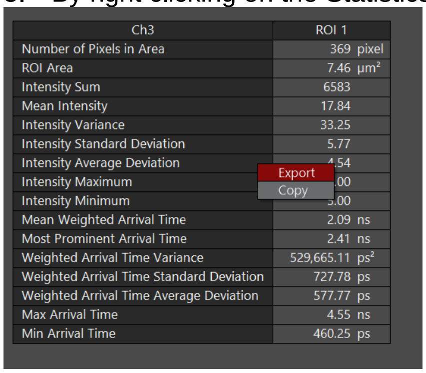
5. By right clicking on the Statistics table you can export your data to csv file.



Ch3	ROI 1
Number of Pixels in Area	369 pixel
ROI Area	7.46 µm²
Intensity Sum	6583
Mean Intensity	17.84
Intensity Variance	33.25
Intensity Standard Deviation	5.77
Intensity Average Deviation	4.54
Intensity Maximum	[illegible].00
Intensity Minimum	[illegible].00
Mean Weighted Arrival Time	2.09 ns
Most Prominent Arrival Time	2.41 ns
Weighted Arrival Time Variance	529,665.11 ps²
Weighted Arrival Time Standard Deviation	727.78 ps
Weighted Arrival Time Average Deviation	577.77 ps
Max Arrival Time	4.55 ns
Min Arrival Time	460.25 ps

System Shut down

1. Save all your images/projects. Close all projects.
2. Switch to Configuration along the top edge.
3. Click on Lasers along the left edge and turn off all the lasers.
4. Close the software. It will take a few minutes to shut down.
5. Turn off the computer or sign out. If another user is coming after you, do not continue with the next steps.

   1. At left from the system, black box, turn off the laser key and flip off the two switches.

Instrument	STELLARIS 8
Serial Number	8300000345
Software Version	LAS X 4.4.0.24861

	Magnification/NA	Immersion	Coverglass	Manufacturer PN	Working Distance
Objectives	1) HC Plan Fluotar 5x/0.15	air/dry	(negligible)	506224	13.7mm
	2) HC PlanApo 10x/0.4 "A"	air/dry	0.17mm ("#1.5") coverglass	506424	2.56mm
	3) HC PlanApo 20x/0.70 "C"	air/dry	0.17mm ("#1.5") coverglass	506513	0.59mm
	4) HC Fluotar L 25x/0.95 W VISIR	water immersion	0.17mm ("#1.5") coverglass	506375	2.4mm
	5) HC PlanApo 40x/1.3 "D"	oil immersion	0.17mm ("#1.5") coverglass	506358	0.24mm
	6) HC PlanApo 63x/1.4 "E"	oil immersion	0.17mm ("#1.5") coverglass	506350	0.14mm

A/B/C/D/E - DIC objective prism "class"

Dichroics 458/514/5 *Dichroics allow for simultaneous*
488/561/6 *acquisitions of multiple*
458/514 *florophores.*
488/594

RT 15/85

NEED TO RECONFIRM EVERYTHING IN RED

Excitation Tunable White Light Laser (WLL) 440-790nm (1nm steps)
405nm diode laser

Detectors 4x Power HyD S (ultra-sensitive detectors for ***All detectors***
1x T-PMT (for transmitted light) *Frequency Range: 10-2600Hz*
Detection Range: 400-850nm

Largest Field Size 3.1mm X 3.1mm (5x objective, 0.75 zoom)

Techniques Available
Simultaneous confocal
Sequential confocal
Lightning mode
Z-stack (13.0mm full; 2.0mm usable)
Tiling (Navigator)
TauSense Modes
Bleaching Experiments

Compatible with:
standard slides (76x26mm)
standard well plates
Petri Dishes (multiple sizes)

Accessories Full incubation setup including stage-top heated incubator (slides, dishes, plates)